

Chapter 1

Latent Space Analysis

The ability of a neural network to generalize provides evidence that the model has learned more than a simple memorization of the training examples. Instead, it must construct internal representations that capture regularities present in the data. These representations emerge progressively through the hidden layers during training and play a central role in the network's ability to solve a task. Understanding their structure is therefore a key step toward understanding how neural networks achieve generalization.

In classical machine learning approaches, practitioners had to manually design data representations in order to make them usable by simple models. Deep learning introduces a paradigm shift: rather than relying on hand-crafted features to meaningfully represent data, it automatically learns these representations. Through the successive activation of hidden layers, neural networks gradually construct representations that become increasingly relevant to the task under consideration. As discussed in Chapter ??, these representations are not meaningful at initialization, they emerge progressively during training as the model optimizes its objective. It is precisely through these learned representations that a trained network is able to map an input to an output, including for examples never encountered during training.

The study of these representations has therefore become a central topic in deep learning research. In particular, researchers seek to understand how information is organized within these representations, what concepts they encode, and how they contribute to the model's behavior. The collection of representations produced by a hidden layer is referred as a latent space.

In this chapter, we first make explicit the working hypothesis that underlie most research on latent space analysis. To do so, we introduce the notions of abstraction, concept, latent space, and semantics. Once the notions of this conceptual framework has been established, we review the main methods that aim to identify and extract concepts from latent representations.

1.1 Hypothesis

In this section, we introduce the main working hypothesis commonly shared in the literature on latent space analysis. It should be noted that this hypothesis is rarely stated explicitly, but rather appears implicitly through the notions

of abstraction, concept, latent space and semantics, used throughout the field. Rather, it should be understood as a set of intuitions and assumptions guiding the study and interpretation of these spaces. The aim of this section is therefore to make this general conceptual framework explicit and to define the main terms associated with it.

This hypothesis aims to explain the generalization ability of neural networks observed after training has been completed, as seen in Section ?? . Formally, generalization is defined as, even for an input $x \in X$ never seen during training, the network is often able to produce an output $\hat{y}$ close to the true output y . This ability implies that the network cannot simply memorize the associations observed during training. Instead, it must capture deeper regularities of the function it approximates. This process requires a form of abstraction: the network must be able to extract the relevant information contained in x and properly combine the elements in order to produce a coherent prediction.

This capacity for abstraction emerges through the construction of meaningful data representations during training. A neural network does not directly map an input x to an output y . Instead, as illustrated in Figure ?? , multiple hidden layers lie between these two endpoints and produce intermediate activations. These activations constitute internal representations of the input. We refer to them as representations because each layer preserves certain properties of x while progressively reorganizing the information into a form better suited to the task. At each hidden layer, the network refines the representation of x , gradually transforming it into a form closer to the desired output. This process can therefore be viewed as a chain of successive transformations:

$$x \rightarrow h^{(1)} \rightarrow h^{(2)} \rightarrow \dots \rightarrow h^{(L-1)} \rightarrow y$$

where each hidden state encodes a progressively more task-relevant representation of the input. These representations are considered abstract because they do not aim to preserve all details of the original input; instead, they retain only the most useful features for prediction.

If a network is able to construct sufficiently meaningful representations of data to ensure generalization, this implies that it can associate a unknown input with structures already encountered during training. These regularities are commonly referred to as concepts. A concept can be understood as a general and abstract representation of a situation, useful for accomplishing a given task. Although this notion remains difficult to formalize rigorously and is still relatively vague in the scientific literature, it intuitively refers to an abstract representation that groups together relevant information for problem-solving. The internal representations of a network can then be interpreted as a combination of several concepts relevant to prediction.

The set of activations of a given layer is referred to as the latent space. This space is not simply an unordered collection of numerical values. Rather, representations are organized according to a specific structure learned during training. The relative position of representations in this space directly influences the behavior of the network. Similar inputs generally produce nearby representations in the latent space, whereas different inputs occupy distinct regions. The latent space thus reflects both the properties of the data, the constraints of the task, and the internal mechanisms learned by the network. Studying these latent spaces is particularly important because it allows us to observe how the network structures information in order to make predictions.

This naturally leads to the question of whether the concepts learned by neural networks are similar to those used by humans when performing comparable tasks. When a human learns a task, they also build abstractions that allow them to generalize to new situations. Researchers therefore seek to determine whether the representational mechanisms observed in neural networks share similarities with those used in human intelligence. When a concept learned by a network corresponds to a human-interpretable concept, it is generally said to have semantic value. However, a network may also learn highly effective concepts for solving a task while remaining difficult for humans to interpret. In such cases, internal representations retain strong functional utility despite low semantic interpretability.

It is important to emphasize that the ideas presented in this section are not yet formally established. The mathematical tools currently available do not provide a complete explanation of the internal functioning of deep neural networks, nor a rigorous proof of the existence of concepts within learned representations. Nevertheless, this perspective currently constitutes the dominant working hypothesis in the community studying latent spaces. Many research works therefore aim to characterize internal representations, identify the concepts present in hidden activations, and better understand how these structures emerge during training.

The remainder of this chapter is devoted to presenting different methods for analyzing latent spaces and extracting the concepts encoded within them.

1.2 Probing

Probing methods aim to determine whether a given human-defined concept is encoded within the internal representations of a neural network. More specifically, they investigate whether the presence or absence of a selected concept can be inferred from embeddings. The underlying assumption is that if a concept can be reliably decoded from these representations, it is likely that the network encodes information related to it in order to perform its task.

A probing analysis begins by identifying a set of interpretable concepts that the network could potentially use to accomplish its task. Each example in the dataset is then annotated to indicate the presence or absence of the studied concept. The data are subsequently projected into the latent space of the pretrained model. From these latent representations, a classifier is trained in a supervised manner to predict the presence of the considered concept. This classifier is referred to as a probe, since its role is to determine whether a concept is present or not. The role of this probe is therefore to assess whether the information associated with the concept is accessible from the network's internal representations.

The classifier used to detect the presence or absence of a concept must remain intentionally simple. Indeed, if the probe has too much capacity, it may reconstruct the target information from complex correlations that do not necessarily correspond to a representation explicitly encoded in the latent space. In such a case, it becomes difficult to determine whether the concept is genuinely represented by the network or merely reconstructed through the expressive power of the classifier. For this reason, probing methods often rely on linear classifiers. A linear separation suggests that the information is easily accessible to the neural

130 network and potentially usable during decision making.

131 If the classifier achieves performance significantly above chance level, this
132 suggests that the latent space contains exploitable information related to the
133 considered concept. However, the fact that a concept can be decoded from latent
134 representations does not necessarily imply that the model actually uses this
135 information to perform its task.

136 To ensure that representations play a causal role in the model’s output, it is
137 possible to perform interventions on the latent representations. An intervention
138 consists of modifying a latent representation in a controlled manner by activating
139 or deactivating a given concept, and then studying its impact on the model’s
140 output. If the resulting change in the model’s output is consistent with the
141 applied perturbation, this suggests that the representation had a causal role.

142 The main limitation of this approach is that the concepts under investigation
143 must be defined in advance before any probing analysis can be performed. This
144 introduces a strong bias in what can be studied. Moreover, annotating a dataset
145 for each concept is extremely time-consuming and requires significant human
146 effort. Interventions in latent space are also difficult to interpret, as it is not
147 always clear what constitutes a meaningful change in the network’s output under
148 a given perturbation.

149 This set of limitations makes probing a highly human-centric approach,
150 strongly dependent on the choice of investigated concepts and on the interpreta-
151 tion of interventions. To overcome this bias, other methods for studying latent
152 spaces have been developed, particularly those aiming to discover concepts in an
153 unsupervised manner.